HOTEL MANAGEMENT SYSTEM

Database Mini Project using Microsoft SQL Server

Introduction

The Hotel Management System is a database management project developed using Microsoft SQL Server. The main purpose of this project is to manage hotel operations efficiently by maintaining records related to customers, room bookings, reservations, room categories, and payments. The system is designed to simplify hotel management activities and improve data organization through structured database design techniques. This project demonstrates practical implementation of SQL Server concepts such as database creation, table relationships, normalization, functions, views, triggers, indexes, transactions, and CRUD operations. The system also ensures proper data consistency and integrity using primary keys and foreign key relationships.

Objectives of the Project

The primary objective of this project is to develop a structured and efficient Hotel Management System capable of handling customer details, room information, reservations, and payment records. Another objective is to demonstrate the practical implementation of SQL Server database concepts learned during the Wipro NGA Training Program conducted by Great Learning. The project also aims to reduce data redundancy through normalization techniques and improve data consistency using relational database concepts. Additionally, the project focuses on implementing CRUD operations, transactions, views, triggers, and functions to simulate real-world hotel management operations.

Technologies Used

This project is developed using Microsoft SQL Server as the primary database management system. SQL Server Management Studio (SSMS) is used for query execution and database administration. The ER Diagram for the project is designed using dbdiagram.io to visually represent the database structure and relationships between tables.

Database Name

The database created for this project is named “HotelManagementSystem”. The database contains all tables, relationships, constraints, functions, views, triggers, indexes, and transactions related to the hotel management process.

Modules of the System

The Hotel Management System consists of several modules that work together to manage hotel operations effectively. The Customer Management module stores customer information such as name, phone number, email, and address. The Room Category module manages different categories of rooms and their pricing information. The Room Management module maintains room details and availability status. The Reservation Management module handles customer bookings, check-in dates, check-out dates, and reservation status. The Payment Management module stores payment details related to customer reservations.

Database Tables

The Customers table stores all customer-related information including CustomerID, FullName, Phone, Email, and Address. CustomerID is used as the primary key to uniquely identify each customer in the system.

The RoomCategory table stores room category information such as category name and room pricing. It contains CategoryID, CategoryName, and PricePerNight attributes. CategoryID acts as the primary key for the table.

The Rooms table stores details of hotel rooms including RoomID, RoomNumber, CategoryID, and room status. The CategoryID in this table is used as a foreign key to establish a relationship with the RoomCategory table.

The Reservations table stores booking and reservation information. It contains ReservationID, CustomerID, RoomID, CheckInDate, CheckOutDate, and ReservationStatus attributes. CustomerID and RoomID are foreign keys that create relationships with the Customers and Rooms tables respectively.

The Payments table stores payment-related information such as PaymentID, ReservationID, Amount, PaymentDate, and PaymentMethod. ReservationID is used as a foreign key to connect payments with reservations.

Primary Keys and Foreign Keys

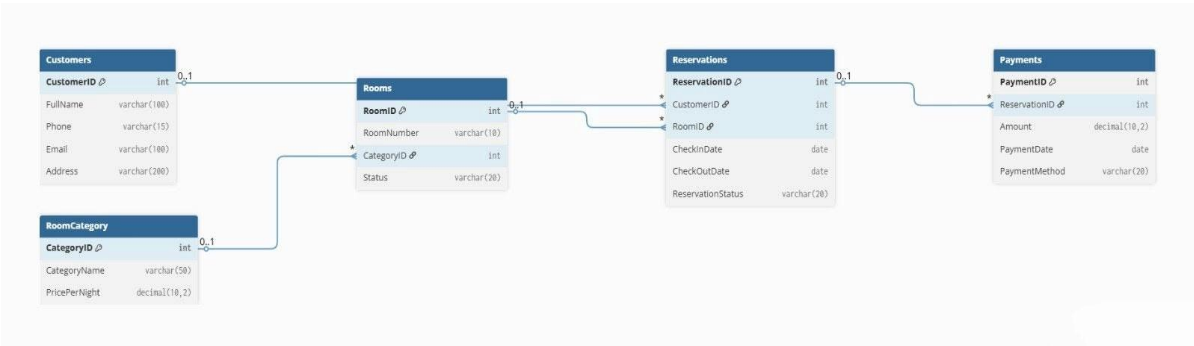
Primary keys are used in the project to uniquely identify each record within a table. CustomerID, CategoryID, RoomID, ReservationID, and PaymentID are used as primary keys in their respective tables. Foreign keys are used to establish relationships between tables and maintain referential integrity. For example, CategoryID in the Rooms table references the RoomCategory table, while CustomerID and RoomID in the Reservations table reference the Customers and Rooms tables respectively. Similarly, ReservationID in the Payments table references the Reservations table.

Relationships Between Tables

The Hotel Management System database contains multiple one-to-many relationships. One room category can contain multiple rooms, which creates a one-to-many relationship between RoomCategory and Rooms. One customer can make multiple reservations, establishing a one-to-many relationship between Customers and Reservations. One room can appear in multiple reservations over time, which forms a one-to-many relationship between Rooms and Reservations. Similarly, one reservation can contain payment details, creating a one-to-many relationship between Reservations and Payments.

ER Diagram Explanation

The ER Diagram represents the database structure of the Hotel Management System visually. It shows entities, attributes, primary keys, foreign keys, and relationships between tables. The diagram helps in understanding how tables are connected and how data flows throughout the system. The cardinality symbols used in the diagram represent the relationships between entities. The symbol "0..1" represents zero or one occurrence, while the symbol "*" represents many occurrences. For example, one customer can have multiple reservations, while each reservation belongs to a single customer.



Normalization

Normalization techniques are used in the project to reduce redundancy and improve data consistency. The database follows the First Normal Form (1NF) by ensuring atomic values and eliminating repeating groups. The Second Normal Form (2NF) is achieved by ensuring that all non-key attributes fully depend on the primary key. The Third Normal Form (3NF) is implemented by removing transitive dependencies. For example, room category details and pricing are stored separately in the RoomCategory table instead of repeatedly storing them in the Rooms table. This improves maintainability and prevents duplicate data storage.

SQL Concepts Used

The project uses several SQL Server concepts to implement database operations effectively. DDL commands such as CREATE, ALTER, and DROP are used to create and manage database objects. DML commands including INSERT, UPDATE, DELETE, and SELECT are used to manipulate data stored in the database. DCL commands such as GRANT and REVOKE are used to control user permissions and database access. TCL commands including COMMIT and ROLLBACK are used to manage transactions and maintain data consistency.

Functions

Functions are implemented in the project to simplify query execution and perform reusable operations. Scalar functions and table-valued functions are used to process data and return calculated or filtered results. Functions help in reducing query complexity and improving reusability within the database system.

Views

Views are created in the project to simplify complex SQL queries and provide abstraction over database tables. Views help users retrieve important information without directly accessing the underlying tables. They also improve readability and enhance security by restricting direct table access.

Triggers

Triggers are implemented to automatically execute actions whenever INSERT, UPDATE, or DELETE operations occur on database tables. Triggers help maintain data consistency, automate database operations, and validate certain conditions during data modification activities.

Indexes

Indexes are used in the project to improve query performance and speed up data retrieval operations. Clustered and non-clustered indexes are implemented to optimize database searching and improve overall system efficiency.

Transactions

Transactions are used to maintain data consistency and ensure reliable execution of database operations. Commands such as BEGIN TRANSACTION, COMMIT, and ROLLBACK are used to manage transactions. Transactions ensure that database changes are either fully completed or completely rolled back in case of failures.

Project Execution Flow

The project execution begins with database creation using DDL scripts. After database and table creation, DML scripts are executed to insert sample data into the tables. Functions, views, triggers, and indexes are then created to enhance database functionality. Transactions are implemented to demonstrate data consistency management. Finally, queries are executed to test the system and generate outputs.

Advantages of the System

The Hotel Management System reduces manual work involved in hotel operations and improves efficiency in managing customer reservations and room details. The use of primary and foreign keys ensures proper data consistency and referential integrity. Normalization reduces data redundancy, while indexes improve query performance. The system also provides better organization and easier maintenance of hotel-related information.

Future Enhancements

The project can be further enhanced by integrating an online booking system, admin dashboard, authentication system, and ASP.NET Core frontend application. Additional modules such as staff management and payment gateway integration can also be added to improve the overall functionality of the system.

Conclusion

The Hotel Management System successfully demonstrates the practical implementation of Microsoft SQL Server concepts through a real-world mini project. The project includes normalization, relationships, functions, views, triggers, indexes, and transactions to create an efficient and scalable hotel reservation system. This project helped in improving the understanding of relational database design, SQL Server operations, and database management techniques learned during the Wipro NGA Training Program conducted by Great Learning.